

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level



MARKING SCHEME



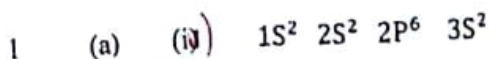
Qn 9

NOVEMBER 2018

CHEMISTRY

6031/3



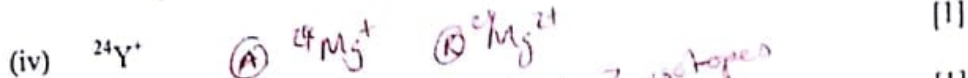


(ii) Average mass of one atom of the element compared to $\frac{1}{12}$ of the mass of one atom of C-12 isotope [1]

$$Ar = \frac{12 \times \text{average mass of one atom of an element}}{\text{mass of one atom of C-12}} \quad (A) \text{ } ^{12}\text{C}$$

(iii)
$$\frac{79.0 \times 24 + 10.0 \times 25 + 11.0 \times 26}{100}$$
 [1]

$$= 24.3$$



(v) Y has (three) different isotopes / Mg has 3 isotopes [1]

(b) (i) the equilibrium constant for the reaction in which water undergoes an acid base reaction with itself; (A) $K_w = [\text{H}^+][\text{OH}^-]$ [1]



$$K_w = [\text{OH}^-][\text{H}^+] = 1 \times 10^{-14}$$
 [1]

$$[\text{H}^+] = 1 \times 10^{-7};$$
 [1]

$$\text{pH} = -\log(10^{-7}) = 7;$$
 [1]

(c) No of moles of NaOH = $\frac{10 \times 0.130}{1000}$ [1]

$$= 0.0013;$$

$$\text{No of moles of HA remaining} = 0.055 - 0.0013/$$

$$= 0.0537;$$
 [1]

$$\text{No of moles of A}^- \text{ present} = 0.025 + 0.0013/$$

$$= 0.0263;$$
 [1]

(A) eq

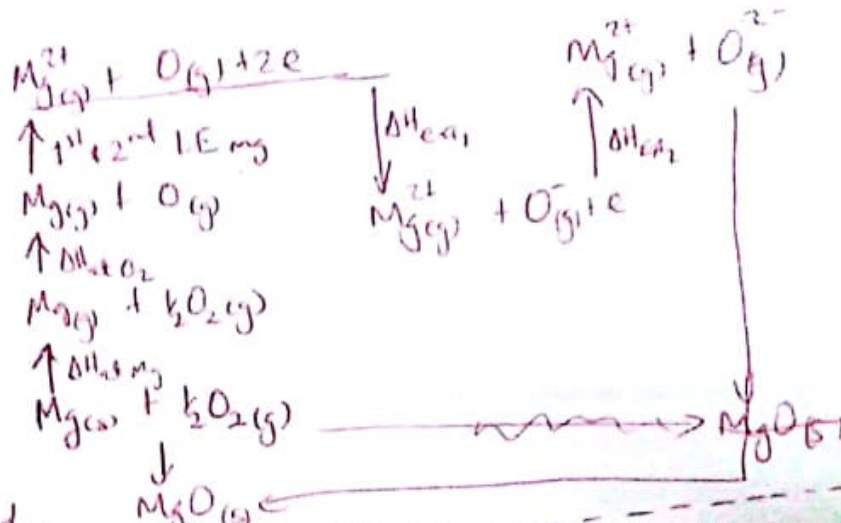
$$\begin{aligned} \text{new } [\text{HA}] &= 0.0537 / 0.11 / = 0.488 \text{ mol/L} \\ \text{new } [\text{A}^-] &= 0.0263 / 0.11 / = 0.239 \text{ mol/L} \end{aligned} \quad (1)$$

$$\text{new pH} = -\log 2.87 \times 10^{-5} = \log \left(\frac{0.488}{0.239} \right) = 4.23 \quad (1)$$

2 (a) (i)

correct cycle [1]
 state symbols [1]
 correct energy terms [1]
 on the cycle

A electron/s
 for one emission, give
 substage mark (1/3)



you can award
 mark for correct
 energy terms/values
 on part (ii) to give
 a total of 4
 the 2 for (i)

(ii) $\Delta H_{2^{nd}EA} = -602 - (150 + 248 + 736 + 1450 - 142 - 3888)$ [1] mark for 248/496/2

$\pm 844 \text{ (kJ mol}^{-1}\text{)}$ [1] mark

(iii) (i) O^{-1} repels the incoming electron;

more energy required to overcome the repulsion; } / 4W

(iv) (ii) it has a high melting point due to (strong electrostatic forces between ions) [1]

(1) thermally stable / resistant to high temp

(b) (i) $S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2(aq)$ ignore s.s but if given must be correct. [1]

(ii) (Comparing experiments 1 and 2)

$[I^-]$ is halved while $[S_2O_8^{2-}]$ is kept constant, rate is also halved. Hence,

order is 1 with respect to I^- / calculation

[1]

(Comparing experiments 2 and 3)

$[S_2O_8^{2-}]$ is doubled, while $[I^-]$ is kept constant, rate is also doubled.

Hence order is 1 with respect to $S_2O_8^{2-}$ / calculation

[1]

Rate = $k [S_2O_8^{2-}] [I^-]$; effect for wrong orders.

[1]

(1 mark for expressing only)

$$(iii) \quad k = \frac{\text{rate}}{[S_2O_8^{2-}][I^-]}$$

$$= \frac{2,2 \times 10^{-4}}{0,080 \times 0,034}$$

$$= 8,1 \times 10^{-2} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}; \quad [1]$$

answer - 1 mark
units - 1 mark

ecf for
incorrect
orders

(iv) Colorimetric/titrimetric/conductometric/

3 (a) (i) Overall reaction: $2H_2 + O_2 \rightarrow 2H_2O$ any explanation of any of these ignore S.S [1]

(ii) $E_{\text{cell}}^{\theta} = 0,83 + 0,40 = +1,23 \text{ (V)}$ ignore units [1]

(iii) - Perpetual supplier of electricity while lead - acid accumulators have to be recharged; [1]

- more efficient

- They are lighter/ lower mass [1]

(iv) Overall reaction is the same ($2H_2 + O_2 \rightarrow 2H_2O$) [1]

(v) Increase the surface area; (so reaction is faster) [1]

(b) Charge passed = $0,03 \times 50 \times 24 \times 60 \times 60$ [1]

$$= 129600 \text{ C};$$

$$n(H_2) = \frac{1 \times c}{2 \times 96500}$$

$$= \frac{129600}{2 \times 96500}$$

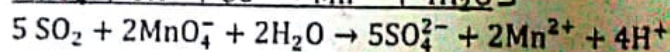
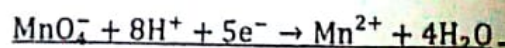
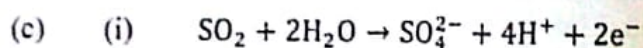
$$= 0,6715 \text{ mol};$$

[1]

$$\text{mass } m(H_2) = 0,6715 \times 2$$

$$= 1,34 \text{ g}$$

[1]



[1]

(ii) $M_r(KMnO_4) = 39,1 + 54,9 + 4(16)$ [1]

5 give mark for Mr if calculations not shown

$$[\text{KMnO}_4] = \frac{3,16}{158} = 0,02 \text{ mol dm}^{-3} \quad [1]$$

$$n(\text{MnO}_4^-) = \frac{7,4}{1000} \times 0,02 = 1,48 \times 10^{-4} \text{ mol} \quad [1]$$

$$\begin{aligned} n(\text{SO}_2) &= \frac{15}{2} \times n(\text{MnO}_4^-) \\ &= \frac{5}{2} \times 1,48 \times 10^{-4} \\ &= 3,7 \times 10^{-4} \text{ mol} \end{aligned} \quad [1]$$

$$M_r(\text{SO}_2) = 32,1 + 2(16) = 64,1$$

$$\begin{aligned} m(\text{SO}_2) &= n \times M_r \\ &= 3,7 \times 10^{-4} \times 64,1 \\ &= 2,37 \times 10^{-2} \text{ g} \end{aligned} \quad [1]$$

AW (b)

$$\begin{aligned} m(\text{H}_2) &= \frac{It M_r}{ZF} \quad [1] \\ &= \frac{0,03 \times 50 \times 24 \times 60 \times 60 \times 2}{2 \times 96500} \quad [1] \\ &= 1,34 \text{ g} \quad [1] \end{aligned}$$

[Total:15]

4 (a) (i) 1 Melting points decrease down the group/graph [1]

increase
Decrease in ionic radii decreases charge densities of ions; Hence
decrease in Forces of attraction between the ions and delocalised
electrons; AW: metallic bond becomes weaker [1]

(ii) 2 Decomposition temperatures increase down the group; [1]

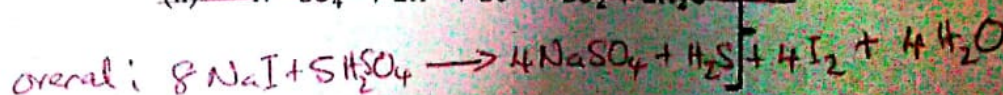
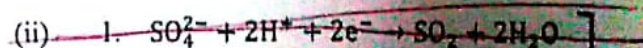
3. Charge density of group II cations decreases; [1]

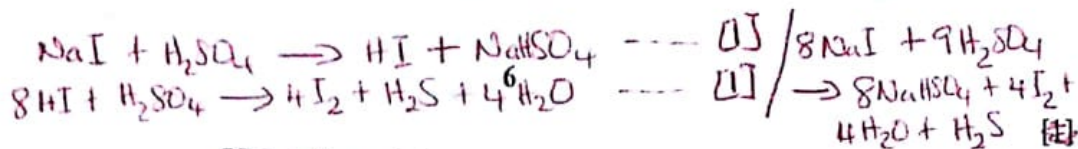
- Polarisation of NO_3^- ion decreases/
polarising power decreases down the group [1]

(b) (i) Iodide/ I^- [1]

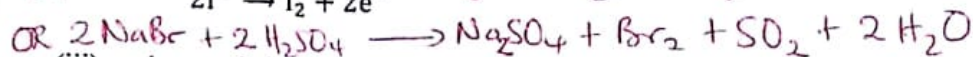
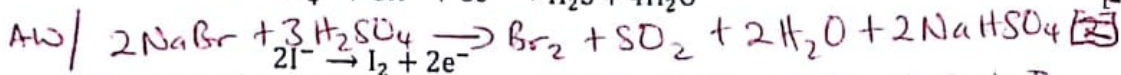
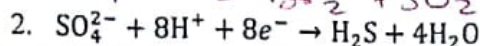
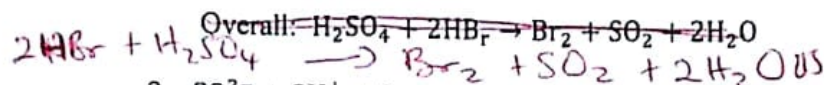
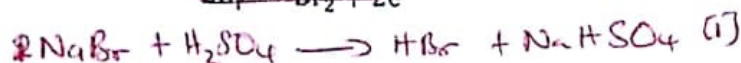
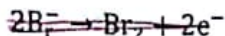
Iodide reduces S from +6 in H_2SO_4 to -2 in H_2S [1]

Bromine reduces sulphur from +6 in H_2SO_4 to +4 in SO_2 . [1]



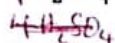
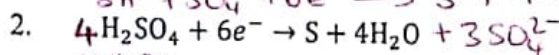


overall eqn
2 marks



overall
eq [2 marks]

(iii) 1. Sulphur



[Total: 15]

5 (a) (i) MgO slightly soluble in water,

CaO readily dissolves in water;

So produces lime water;

} / MgO is soluble
+ then CaO
A reacts slowly/fast



(ii) burn readily to produce bright / coloured flames; Explode

(b) (i) - Volatility decreases from F₂ to I₂; Explode

- Strength of Van der Waals; increases with increase in electron cloud/molecular size mass

(ii)

Hydride	Bond Energy/k Jmol ⁻¹
H - F	562
H - Cl	431
H - Br	366
H - I	299

- Thermal stability decreases from HF to HI; Explode

- HX becomes easier to break down the group as bond energy decreases / bond becomes longer & weaker

(c) - Weigh out accurately a known mass x g of LoSalt;

for Siln:

Handwritten notes:
 $n(H) = 1$
 $n(C) = 1$
 $n(H) = 1$
 $n(C) = 1$

Make a copy of this and write highlighted notes.

1. Name the compound of this solution with light rays.

Added.

First preparation solution in solution.

Continuous treatment with positive results with 100%.



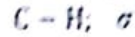
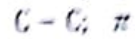
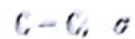
(c) Structural isomers:



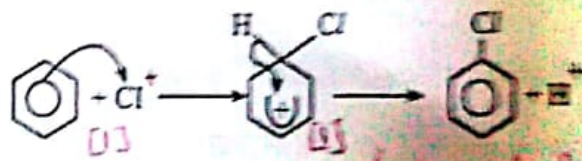
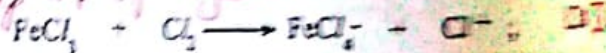
(iii) distillation.

(iv) (partial) oxidation.

(d) (i)



(ii)

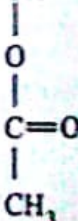
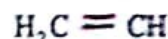


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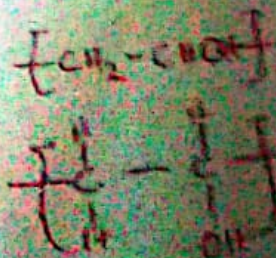
(a)

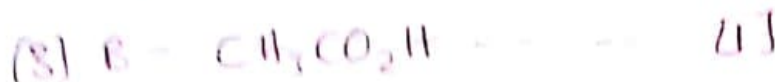
(i)

1.

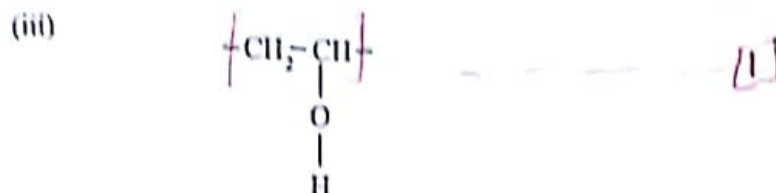


2.



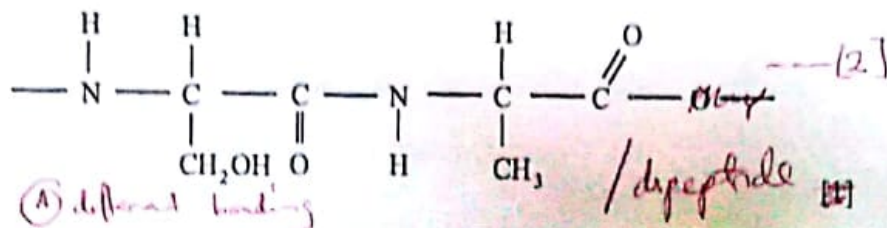


(ii) $\text{C}=\text{C}$ double bond/ $\text{C}=\text{C}$ alkene; ester; (4) structure [2]

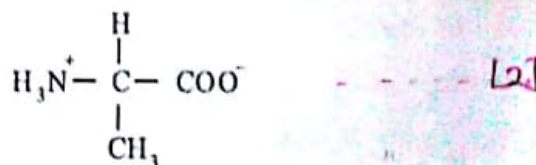


(iv) NaBH_4 in methanol/ LiAlH_4 in dry ether/ $\text{H}_2, \text{Ni}^{2+}$ catalyst 140-150°C [2]
correct reagent 1
condition 1

(b) (i) 1.



2.



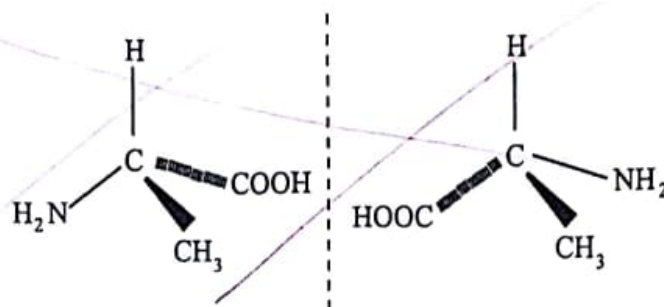
(ii) In acidic solution NH_2/COO^- group accepts a proton/accepts [1]

In alkaline solution $\text{COOH}/\text{NH}_3^+$ group donates a proton/donates [1]

(iii) Both W and Z/ having chiral centre } [1]
Carbon atom surrounded by four different substituents;

1. Optical

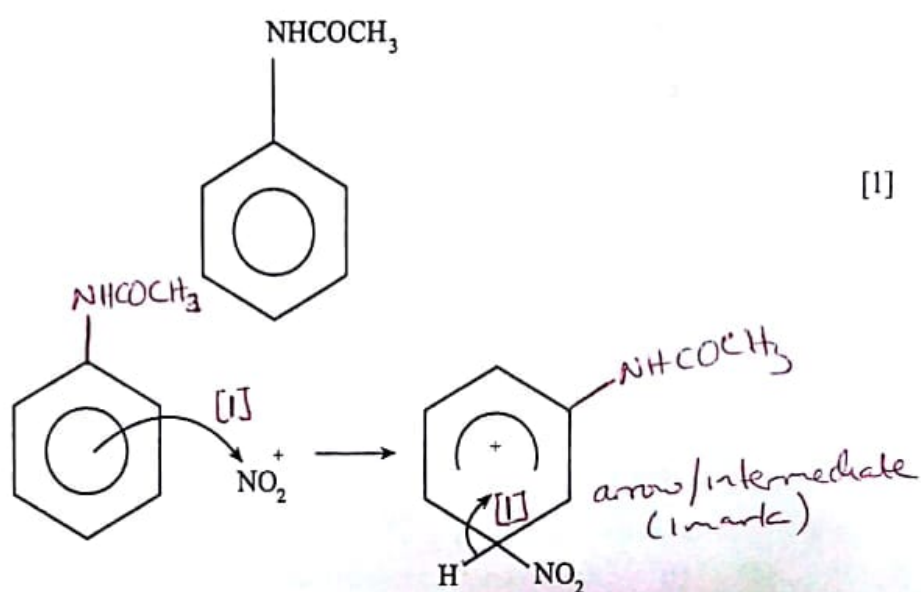
[1]



[1]

[Total:15]

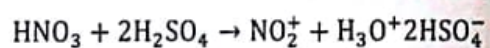
8 (a) (i)



[1]

(ii) electrophilic substitution;

[1]



[1]

(iii) (alkaline) hydrolysis;

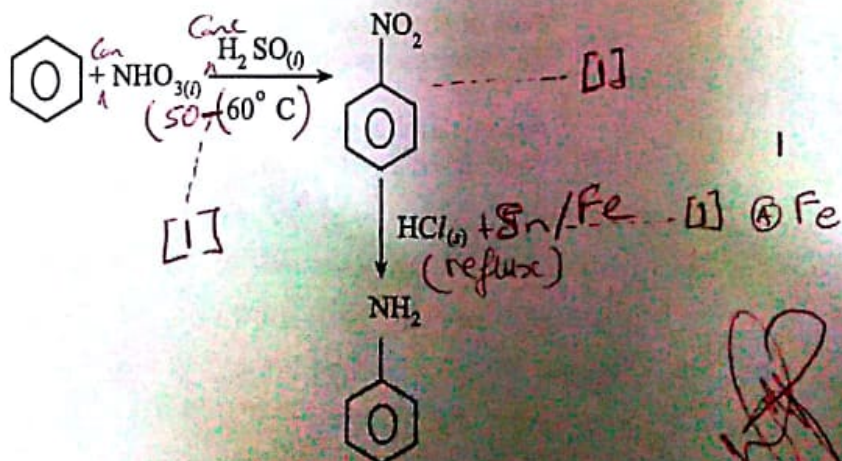
② acid hydrolysis

[1]

(iv) 12. NaOH(aq), warm; / reflux / heat / boil

[1]

(b)



- (c) (i) carbonyl group / ketone or aldehyde / $-CH=O$ / $C=O$ [1]
 not an aldehyde; reacts with 2,4-DNP [1]
 not an alcohol; not an aldehyde / will be a ketone [1]
 should be a ketone; no reaction with $K_2Cr_2O_7/H^+$ [1]
 (ii) $CH_3 - CO - CH_2 - CH_3$; (A) displayed [1]
 [Total: 15]

9

- (a) (i) Extract DNA / collect cell sample from DNA material [1]
 (ii) Use restriction enzyme / cut DNA using restriction enzyme [1]
 (iii) Use polymerase chain reaction / PCR amplification [1]
 (iv) Place samples on agarose gel / ~~separate~~ separate DNA fragments [1]
 and by electrophoresis [1]
 (v) Carry out electrophoresis [1]
 (vi) Transfer DNA onto a paper / nylon membrane [1]
 (vii) Label with radioactive isotope / add radioactive probe [1]
 (viii) Measure radioactivity / set up an X-ray film [1]
 (i) Phosphorus - 32 (or) compare with known X-ray film [1]

Any 2 correct
for 1 mark
ignore order

- (b) (i) It makes the DNA fragments/bands visible / enables location of DNA fragment [1]
 (ii) Size of the ions / charge on the ions / voltage / temperature / pH / pressure of mass spectrometry / resistance of side chains or stage of molecule / charge density / Any two [2]
 (d) (i) K is Lysine / It moved to the negative electrode. The side chain is positively charged. [1]
 L is alanine. It does not move because the pH of the solution is the same as its isoelectric point / exists as a zwitterion / has no net charge [1]
 M is glutamic acid [1]
 N is aspartic acid [1]

Substitute mark
for 2 correct
[1 mark]